Justin Verrick

INFO-C 451

3/2/2026

Midterm Project

System Implementation within SDLC: Car Rental System

For this project, the Car Rental System follows SDLC phases. The SDLC phases are planning, analysis, design, development, testing, implementation, and maintenance. System implementation will take place after development and testing are done.

During the planning phase of SDLC, the organization will identify the need to replace their outdated manual booking process with an improved full digital system. Requirements will be gathered in the analysis phase. These requirements are customer reservations, vehicle tracking, payment processing, and reporting. The design phase of SDLC will identify the database structure, system architecture, and user interface. The development phase of SDLC will focus on building the application and integrating system components.

Implementation phase of SDLC will occur after system testing confirms the system meets requirements. Implementation is preparing the environment, transferring existing rental and customer data, training employees, and deploying the system based on the selected installation strategy. Implementation phase is not just turning the system on. It involves risk management, data conversion, user training, and planning to make sure that there's little trouble with the daily operations. Last is the maintenance phase of SDLC which follows the implementation phases. In the maintenance phase updates, bug fixes, and performance improvements happen.

Chosen Development Methodology: Hybrid Approach

For the Car Rental Management System, I chose a hybrid development approach that uses elements from both Waterfall and Agile methods. I believe this approach makes the most sense to the car rental system because the organization needs structured planning but will still need some flexibility during development.

The earlier phases of the project, like planning, analysis, and high-level design, will follow a traditional Waterfall structure. This will allow the organization to clearly define system requirements, timelines, budget expectations, and overall goals before development happens. Since this system is going to replace an outdated manual process, it is very important to have strong documentation and a clear map from the start.

During the development and testing phases of SDLC, Agile principles will be used. System features like reservation validation, payment processing, and reporting can be developed and tested in smaller sections. This will allow feedback from users and stakeholders to be gathered early, and adjustments can be made without restarting the project. It will also help reduce risk because the issues can be found early and corrected in much smaller cycles.

Using this hybrid approach will provide structure while still allowing room for improvement and refinement for the system. It will make sure that the project stays organized and on schedule but will also keep it flexible if changes are needed. For car rental management platform system, having a balance between control and flexibility is important to make sure that successful implementation happens.

Planning Objectives and Implementation Strategies

The primary objective of this project is to effectively replace the outdated manual car rental system with a centralized and digital Car Rental System. The new system must improve

efficiency, reduce human error, and provide tracking of reservations, vehicle availability, and payments.

One planning objective is to define system requirements before implementation starts. Doing this will make sure that features like reservation management, customer records, vehicle inventory tracking, and payment processing will function right from the start. Another objective is to set a reasonable timeline and distribute resources. This will include developers, testers, and trainers.

Risk reduction is an important planning strategy. Identifying potential technical issues, data migration risks, and employee struggle to change. Early communication with staff and proper training will be scheduled to support user adoption.

A structured communication plan will be implemented to keep stakeholders informed of progress, milestones, and any issues that happen. Regular meetings and progress reviews will make sure that the project remains aligned with business goals.

The planning phase will focus on preparation, coordination, and reducing errors to daily operations during the system's deployment.

Risk Assessment and Mitigation Plan

Implementing a new Car Rental System involves a few potential risks that must be detected early and managed correctly. A structured risk assessment will help reduce the chance of major interruptions during implementation phase.

One major risk is data migration errors happening. Since the organization is moving away from a manual and outdated system, customer records and rental history could contain inconsistencies and/or missing information. If the data is transferred inaccurately, it could have an impact on reservations and reporting. To mitigate this risk, data will be reviewed and cleaned

before migration happens. Test migrations will be performed in a staging environment to verify data accuracy.

Another risk is system downtime during the deployment phase. If the system is inaccessible for an extended period, it could interrupt rental operations and affect customers in a negative way. To reduce this risk, implementation will be scheduled during off hours, and backup processes will be prepared.

Employee resistance to change is a possible risk. Staff members who are used to the old process may struggle to change to the new system. This risk will be addressed through training sessions, user guides, and demonstrations. Providing good support during the transition period will increase confidence and reduce employee frustration.

There's a technical risk related to integration issues, such as payment processing failures or database performance problems. This will be mitigated through thorough system testing, including unit testing, integration testing, and user acceptance testing before deployment. By identifying risks early and preparing mitigation strategies, the organization can reduce uncertainty and improve the chances of good implementation.

Risk Assessment Matrix

Risk	**Impact**	**Chances of Happening**	**Mitigation**
Data Migration Errors	High	Medium	Clean and Validate Data Before Migration.

System Downtime During Deployment	Medium	Medium	Deploy during Off-Hours and Prepare Backups.
Employee Resistance to Change	Medium	High	Provide Training and Support.
Payment or Integration Failures	High	Low	Perform Integration and System Testing.

Project Timeline

Weeks 1–2: Planning Phase

During the planning phase, project objectives are finalized, stakeholders are identified, and system scope is clearly defined. A preliminary risk assessment is conducted, and resource allocation is established.

Weeks 3–4: Requirements Analysis

Detailed system requirements are gathered, including reservation management, vehicle tracking, payment processing, and reporting features. Meetings with stakeholders make sure that all the business needs are documented.

Weeks 5–6: System Design

The database structure, system architecture, and user interface design are developed. Security considerations and integration requirements are defined during this phase.

Weeks 7–10: Development Phase

Core system features are built, including reservation processing, customer management, inventory tracking, and payment integration. Development follows cycles to allow early testing of features.

Weeks 11–12: Testing Phase

System testing is conducted, including unit testing, integration testing, and user acceptance testing. Any defects found are corrected before deployment approval.

Week 13: Data Migration Preparation

Existing rental records and customer data are cleaned, validated, and prepared for migration. A test migration is performed in a staging environment to verify the data is accurate.

Week 14: Implementation / Deployment

The system is deployed according to the decided installation strategy. Employees receive training, and support is provided during the transition period.

Weeks 15–16: Post-Implementation Support and Evaluation

System performance is monitored, user feedback is collected, and any issues are resolved. A post-implementation review is conducted to evaluate project success.

Gantt Chart

Phase	**What Will Happen**	**Timeline**
Planning	During the planning phase, project objectives are finalized, stakeholders are identified, and system scope is clearly defined. A preliminary risk assessment is conducted, and resource allocation is established.	Weeks 1-2
Requirements Analysis	Detailed system requirements are gathered, including reservation management, vehicle tracking, payment processing, and reporting features. Meetings with	Weeks 3-4

	stakeholders make sure that all the business needs are documented.	
System Design	The database structure, system architecture, and user interface design are developed. Security considerations and integration requirements are defined during this phase.	Weeks 5-6
Development	Core system features are built, including reservation processing, customer management, inventory tracking, and payment integration. Development follows cycles to allow early testing of features.	Weeks 7-10
Testing	System testing is conducted, including unit testing, integration testing, and user acceptance testing. Any defects found are corrected before deployment approval.	Weeks 11-12
Data Migration Preparation	Existing rental records and customer data are cleaned, validated, and prepared for migration. A test migration is performed in a staging environment to verify the data is accurate.	Week 13
Implementation	The system is deployed according to the decided installation strategy. Employees receive training, and support is provided during the transition period.	Week 14
Post-Implementation Support	System performance is monitored, user feedback is collected, and any issues are resolved. A post-implementation review is conducted to evaluate project success.	Weeks 15-16

Installation Approach: Phased

For the Car Rental System, I chose a phased installation approach. I believe this is the best method because it reduces risk and allows the organization to gradually transition. Gradually transitioning is better than all at once because that would cause many problems.

With a phased approach, different system components are implemented in stages rather than all at once. For example, the organization could first implement customer management and vehicle tracking features. After those components are stable and employees are comfortable using them, reservation processing and payment integration can be introduced.

This approach will lower the risk of major disruption. If an issue occurs in one phase, it can be fixed before moving to the next stage. This is important because the organization is moving away from a manual process. A direct cutover would create a higher risk because the entire system would go live at once, leaving very little room for adjustment.

A phased implementation supports employee training. Staff can learn the system slowly instead of being overwhelmed with all features at once. This will improve user adoption and reduce their resistance to change.

Phased implementation will take slightly longer than a direct cutover, but it will provide better control, lower risk, and a smoother transition. For a car rental management system that affects daily operations, this approach provides a balanced and practical solution.

Installation Schedule

The installation schedule for the Car Rental System will follow a phased transition approach to reduce disruption and risk. The rollout will occur in structured stages over a short period of time after final testing is done.

Phase 1: Environment Preparation (Beginning of Week 14)

The production environment will be configured, including servers, databases, and network settings. Backup procedures will be verified to make sure that the data is protected in case of unexpected issues. Final system checks will be performed before deployment begins.

Phase 2: Initial Feature Deployment (Mid-Week 14)

Core features like customer management and vehicle inventory tracking will be activated. Employees will begin using these features while the remaining modules remain inactive. Technical support will be available during this transition period to direct any issues.

Phase 3: Reservation and Payment Module Activation (End of Week 14)

After confirming stability in the first phase, reservation processing and payment integration will be deployed. System performance will be monitored to make sure that transactions are processed right.

Phase 4: Employee Training and Support (Week 15)

Hands-on training sessions will happen to make sure that all staff members understand how to use the system. User guides and support documentation will be provided. Feedback from employees will be collected to identify any usability issues.

Phase 5: System Stabilization and Monitoring (Weeks 15–16)

The system will be continuously monitored for performance, security, and data accuracy. Any issues will be resolved promptly. A final review will happen to confirm the system is fully operational and that it meets business requirements.

This phased installation schedule makes sure that the organization transitions from the old manual process to the new system without overwhelming employees or disrupting daily operations.

Data Migration Plan: Extract, Transform, Load

A big part of implementing the new Car Rental System is transferring existing data from the old manual and outdated system into the new database. To manage this process properly, Extract, Transform, Load approach will be used.

Extract

The first step is extracting the existing data. This includes customer records, rental history, vehicle information, and payment records. Since the current system is manual or partially digital, data may exist in spreadsheets, paper records, or older databases. All relevant data sources will be collected. Before extraction is finalized, data backups will be created to prevent any accidental loss. This will make sure that the original records are saved in case errors occur during migration.

Transform

After extraction, the data must be reviewed and cleaned. This is an important step because older systems often contain duplicate entries, missing fields, or formatting inconsistencies.

During the transformation phase:

Duplicate customer records will be identified and merged. Missing or incomplete data will be flagged for review. Data formats like dates and payment will be standardized. Old data fields will be mapped to the new system's database structure.

Load

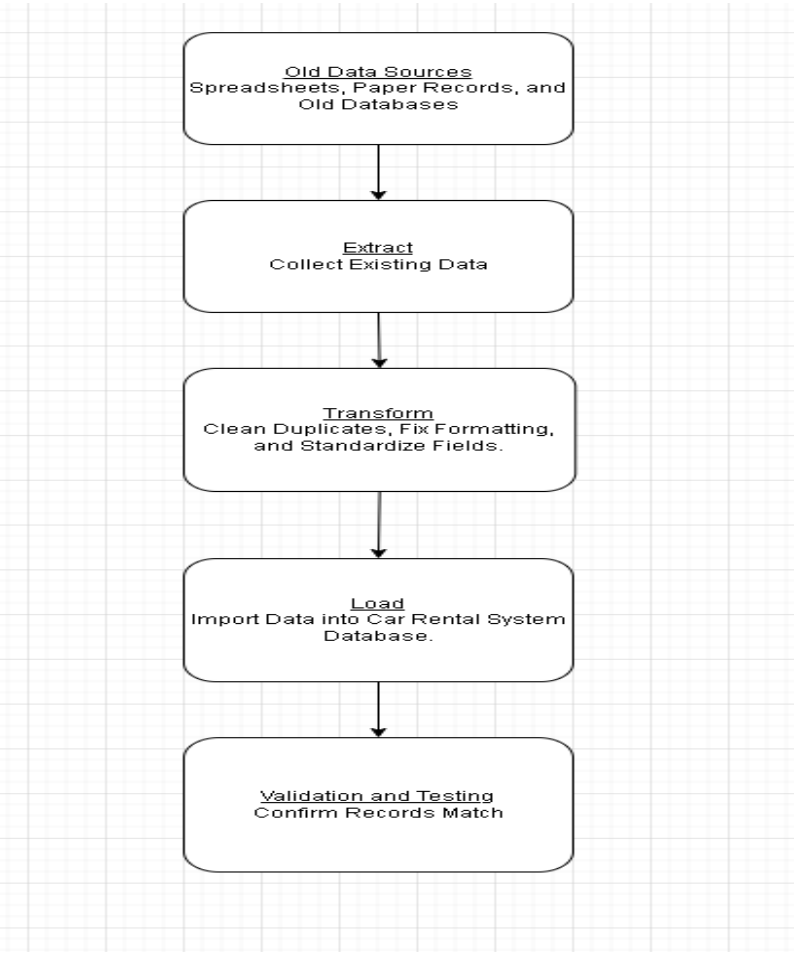
Once the data has been cleaned and properly formatted, it will be loaded into the new Car Rental System database. A test load will first be performed in a staging environment to confirm everything transfers right.

After verifying the test migration, the final data load will happen during a scheduled maintenance window to minimize disruption. Once loading is complete, validation checks will be performed to confirm that records are accurate and complete.

Post-Migration Validation

After the full migration, the system will be reviewed to confirm that customer accounts display correct rental history. Confirm vehicle inventory data matches physical inventory. Confirm payment records are stored right. Any inconsistencies will be corrected immediately before full system operation begins.

ETL Diagram



Test Plan

Before full deployment of the Car Rental System, a structured test plan will be implemented to make sure the system performs correctly, securely, and reliably. Testing is an important step to confirm that the system meets business requirements and does not introduce new operational issues.

Scope

The scope of testing includes all core system functions such as customer registration, vehicle inventory tracking, reservation processing, payment integration, and reporting features. Both functional and non-functional features of the system will be tested. Functional testing will verify that each feature works, while non-functional testing will evaluate performance, security, and system stability.

Objectives

The primary objective of testing is to confirm that the system meets the defined requirements gathered during the analysis phase. Other objectives are making sure that reservation transactions process correctly, verifying payment transactions are secure and accurate, confirming customer and vehicle data is stored and retrieved, identifying defects before implementation, ensuring the system can handle user volume without performance issues.

Testing Tools

A combination of manual and automated testing tools will be used. Manual testing will allow testers to simulate real-world user actions such as booking a vehicle or processing a payment. Automated testing tools may be used to repeatedly test login processes, reservation workflows, and database queries.

Roles and Responsibilities

Testing responsibilities will be divided among team members to make sure that there's solid coverage. Developers will conduct unit testing and correct found defects. Testers will perform system testing and document issues. Project manager will manage testing progression and make sure that milestones are met. End users will participate in user acceptance testing to confirm the system meets operational needs.

Test Cases

Test Case 1: Login Validation with Valid Credentials

Test Case ID: TC-LOGIN-01

Test Type: Functional

Test Goal: Confirm that the user can log in with correct credentials.

Preconditions: User account already exists in the system, can be customer or admin/staff. System is online and connected to the database..

Test Data: Username: testCustomer1, Password: Pass12345

Steps: 1. Navigate to the login page. 2. Enter username and password. 3. Click the "Login" button.

Results: The system logs the user in. The user is redirected to the correct dashboard, customer or admin/staff dashboard. The session remains active until logout.

Test Case 2: Login Validation with Invalid Password

Test Case ID: TC-LOGIN-02

Test Type: Validation and Security

Test Goal: Confirm that the system blocks access when login info is incorrect.

Preconditions: User account already exists in the system.

Test Data: Username: testCustomer2, Password: Pass54321

Steps: 1. Navigate to the login page. 2. Enter valid username and invalid password. 3. Click the "Login" button.

Results: The system denies the login attempt. The user stays on the login page. A message appears "Invalid Username or Password". The system doesn't reveal which field was incorrect for security purposes.

Test Plan Table

Test Case ID	Test Description	Input	Expected Results
TC-LOGIN-01	Login with Valid Credentials	Username: testCustomer1 Password: Pass12345	User Logs in and is Redirected to Dashboard.
TC-LOGIN-02	Login with Invalid Password	Username: testCustomer2 Password: Pass54321	System Denies Access and Shows Error Message.

Debugging and Error Handling

Debugging and proper error handling are important to make sure the Car Rental System operates dependably. Even with good planning and testing, issues can still happen once actual users start using the system.

When a defect is found, the first step is to reproduce the issue. This helps confirm whether the problem is repeatable. Developers will then review system logs, error messages, and database activity to determine the cause. The issue will be documented with the steps needed to recreate it so it can be fixed and tested again.

Bugs will be prioritized based on severity. Critical issues like payment processing failures will be addressed quickly because they affect business operations. Smaller issues like display or formatting problems will still be corrected but not as fast.

The system will use clear and user-friendly error messages to prevent confusion. For example, if a customer enters incorrect login information or selects unavailable rental dates, the system will display a message instead of a technical error. Regression testing will make sure that the changes do not create new problems somewhere else in the system.

Automated Testing Tools

Automated testing tools can help make testing the Car Rental System faster and more consistent. Instead of repeating the same manual tests, these tools can run test automatically and help find problems quicker.

One useful tool is Selenium. It can test the web interface of the system. It can simulate user actions like logging in, selecting vehicles, or making reservations. This helps confirm that important features are working right.

Another tool is Junit. Developers can use it to test smaller parts of the system code. It can test things like reservation validation or payment calculations to make sure the system processes data right.

Postman can be used to test API connections if the system communicates with external services such as payment processing. It helps verify that requests and responses between systems are working. Automated testing does not replace manual testing, but it helps improve efficiency and makes it easier to run important tests repeatedly during development.

Data Quality Issues

When data is transferred from an older system into a new one there is always some risk with data quality. Since the previous car rental process relied on manual records, some information may be inconsistent, incomplete, or just not there.

One common issue is double customer records. The same customer may appear many times with different information. These duplicates will be merged so each customer has a single record. Another issue could be missing or incomplete data, such as missing phone numbers or email addresses. These records will be corrected if possible before the migration.

There could be formatting inconsistencies, especially with dates, phone numbers, or payment information. All these types of values will be consistent, so they match the format used in the new system. A data cleansing process will be performed before migration. This includes removing duplicates, correcting formatting errors, and verifying important records.

Conversion Method

For the Car Rental System, a phased conversion method will be the best option to use. This approach fits well with the phased installation planned for the system implementation. Instead of switching everything at once, the transition will happen in stages.

Different parts of the system will begin using the new database and software in stages. For example, customer records and vehicle inventory may be transferred first. After those parts are confirmed to work, reservation processing and payment functions can be moved to the new system.

This method reduces the risk of problems. If an issue happens during one stage of the conversion, it can be corrected before the remaining system sections are moved over. This helps prevent disruptions to operations.

Another advantage of phased conversion is that employees can adjust to the new system slowly. Staff members will have time to learn about the new processes and become comfortable with the system before more features are added.

A phased conversion may take longer than switching everything at once, but it will provide better control. For an organization moving from a manual system to a fully digital Car Rental System, this approach helps make sure that a smoother and manageable implementation happens.

Testing and Validation Steps

After the data has been transferred into the new Car Rental System, many validation steps will be performed to make sure the data is transferred and that it is accurate. Even when migration is planned, errors can still happen.

The first step will be to compare records between the old system and the new database. Random samples of customer records, rental history, and vehicle inventory will be reviewed to confirm that the information matches the original data.

System queries and reports will be used to verify record counts. For example, the number of customer accounts, vehicle records, and reservation entries in the new system should match the totals from the other system.

Functional testing with the migrated data will occur. Employees will perform simple tasks like looking up customers, creating reservations, and viewing rental history to confirm that the data behaves right.

Any errors or inconsistencies discovered during validation will be fixed before the system becomes operational. These checks help make sure that the data in the new Car Rental System is correct, reliable, and ready.

System Implementation

All the components in this project work together to support the successful implementation of the Car Rental System. The SDLC provides the overall structure for planning, designing, developing, testing, and deploying the system.

The hybrid development methodology allows the project to have clear planning while still leaving room for adjustments during the development and testing phases. The project timeline, planning strategies, and risk assessment help keep the project organized and reduce the risk of problems.

Data migration planning, including the ETL process and data cleansing steps, helps make sure that the customer records, vehicle data, and rental history are transferred right. The phased conversion method helps reduce risk by allowing the system to be introduced slowly.

Testing is an important part of the implementation process. The test plan, test cases, debugging strategies, and automated testing tools help find issues early and confirm that the system works.

These components help make sure that the new Car Rental System is implemented smoothly and can support the organization's operations.

Key Success Factors and Post-Implementation Review Criteria

Many factors will determine whether the Car Rental System implementation is successful or not. One important factor is proper planning and communication through the entire project. Clear requirements, a structured timeline, and regular communication help keep the project organized.

Another important factor is accurate data migration. Customer records, vehicle inventory, and rental history must be transferred correctly into the new system. The data cleansing and validation steps help reduce errors and make sure that the information is reliable.

User training is important for success. Employees need to understand how to use the new system. Training and support will be provided during the transition period.

A post-implementation review will be conducted to gauge system performance. This review will examine whether the system meets business requirements and improves efficiency. Feedback from employees will be used to identify any areas that may need improvement still.

Lessons Learned and Continuous Improvement

This project shows how important planning and communication are when implementing a new system. One lesson I learned is that taking time to define requirements and plan the process can help prevent problems later. Identifying potential risks quickly and having strategies to handle them will make the transition much easier.

Another lesson I learned is how important testing and validation are. The test plan, test cases, and data validation steps make sure the system works right before it becomes operational. Without these important steps, small errors could cause large problems.

Continuous improvement is important for future system implementations. After the Car Rental System is online, feedback from employees and system performance reports can be used

to identify areas that still need improvement. Updates, bug fixes, and system enhancements can then be applied over time.